

Weathering the Music Charts

*A Study on the Influence of Weather on Music Chart Trends

Sruthi Satyavarapu
Computer Science
University of the Pacific
Stockton, USA
s_satyavarapu@u.pacific.edu

Deepika Jakati
Computer Science
University of the Pacific
Stockton, USA
d_jakati@u.pacific.edu

Abstract—This project investigates the impact of weather conditions on music charts and if it plays a role. By combining music chart data with weather patterns, the study examines how various weather factors such as temperature, weather conditions affect the popularity of songs in various locations throughout the world. The analysis employs exploratory data techniques to identify correlations between weather conditions and charts.

Index Terms—weather, music charts, EDA, data science, music

I. INTRODUCTION

In today’s digital age, music is more accessible than ever, and global listening rates have skyrocketed. On average, listeners spend 20 hours a week listening to music per week [1]excluding physical media—making streaming services the dominant mode of consumption. These platforms rely heavily on algorithmic recommendation systems to suggest new songs to users [2]. However, recommending music is a complex challenge due to the multi-dimensional nature of musical taste. Research has shown that emotional tone and lyrical content [3] significantly influence user preferences, improving the effectiveness of recommendation models. Furthermore, societal and cultural backgrounds [4] also play a crucial role in shaping individual and regional music preferences.

The vast amounts of behavioral data exposed by these platforms—through APIs and trend analyses—offer researchers a unique opportunity to study patterns that were once considered too abstract or subjective. By connecting listening behavior (such as genre or tempo preferences) to external variables like weather, we can begin to explore whether music consumption exhibits measurable trends influenced by environmental contexts. In this way, streaming platforms serve not only as entertainment services but also as powerful tools for deriving patterns on a global scale.

With the integration of artificial intelligence and recommendation algorithms [5] [6], music platforms have moved beyond simple playback to deliver personalized listening experiences tailored to user behavior. Features such as mood-based playlists, real-time listening analytics, and smart suggestions powered by machine learning have turned music consumption into a form of self-expression closely tied to lifestyle and emotional context.

A. Data Sources

1) *Weather Data*: We explored multiple weather data providers such as NOAA and WeatherStack, but found that accessing historical weather data often required paid subscriptions. Eventually, we chose the OpenWeather API due to its accessible free tier and support for better detailed coverage.

To collect the necessary data, we initially set up a manual process using a Python script to query the OpenWeather API for specific cities. This script fetched daily weather metrics like temperature, condition, precipitation, and humidity. As the scope of the project expanded, we automated the process to handle all required cities consistently, enabling scalable and continuous data collection.

2) *Music Data*: Music chart data reflects the popularity of songs based on various metrics such as streaming counts, radio play, and sales. By analyzing these charts, we aimed to uncover patterns in music popularity across different regions and cultures. For our project, the focus was on tracking the performance of songs at the city level to understand local trends and preferences.

We initially explored multiple platforms for music chart data, including YouTube, Spotify, and Apple Music. YouTube offered limited chart data focused primarily on global or national trends, lacking granularity for city-level analysis. Spotify Charts provided useful data but were typically organized by country rather than individual cities, limiting the scope of localized comparisons.

Apple Music, however, offered city-level charts, which were ideal for capturing detailed local music trends. Access to Apple’s data through their official API, however, requires a paid developer account, which was not feasible for our project. As a result, we developed a solution using Python and Selenium for browser automation, enabling us to scrape city-level chart data directly from the Apple Music website. This process was conducted exclusively for academic and research purposes.

To ensure a diverse and comprehensive dataset, we collected charts from 103 cities across various continents, including North America, Europe, Asia, Africa, South America, and Oceania. This global approach allowed us to capture regional differences in music preferences and trends, providing a broader view of how cultural factors might influence music

popularity. The cities were selected to reflect a wide range of geographical, cultural, and demographic diversity, helping to ensure that our analysis accounted for various cultural influences on the music charts.

B. Challenges

During the course of this project, we encountered several challenges that required careful planning and adaptation to overcome:

Large Volume of Data and File Management: As we gathered music chart data for 103 cities over an extended period, the volume of data quickly became overwhelming. Each city required multiple files to store daily rankings, which led to issues with organizing and managing a large number of files.

Syncing and Communication Between Team Members: As the project was a collaborative effort between two people, effective communication and synchronization were key to maintaining progress. Managing file access, tracking updates, and ensuring that both team members were aligned with the latest changes was a logistical challenge. GitHub served as a solution, enabling us to collaborate in real time and maintain version control, making it easier to work together, share files, and avoid conflicts.

OpenWeather API Limitations: Initially, we planned to collect historical weather data using the OpenWeather API. However, historical data was a paid service, which was beyond our project's scope. As a result, we shifted our approach and started collecting daily weather data instead of relying on historical datasets. While this solution was practical, it meant that we had to focus on capturing real-time weather data on a daily basis rather than analyzing historical trends. It would have been more ideal if we had access to an entire year's worth of weather data to provide deeper insights into long-term trends and their correlation with music chart performance.

Throughout the project, we encountered several data-related issues that required such workarounds unfortunately.

Limited City-Level Chart Data from Spotify: The Spotify API only offered music chart data by country, which was insufficient for our analysis, as we needed data on a city level to account for the highly localized nature of weather patterns. To address this, we used web scraping techniques with Selenium and Python to collect city-level chart data from websites like Apple Music, which provided top city charts. This approach, while effective, was time-consuming and required careful monitoring to ensure the data's accuracy.

Raw Data and Lack of Standardization: The music chart data we collected was raw, containing only song names and artist information, with no standardized identifiers. This lack of consistency made it difficult to match songs across sources. To resolve this, we relied on Spotify to standardize song names and obtain unique identifiers such as Spotify IDs.

Legal and Logistical Challenges with Music Files and Audio Analysis: We had initially planned to use Spotify audio features via its API, however Spotify had deprecated that API. We had to explore alternative APIs and metadata providers for legal audio previews for analysis. While this allowed us to access

the necessary audio features legally, the previews were limited in duration and did not always fully represent the entire song.

Limited Features from Alternative Data Sources: After losing access to Spotify's audio features API, we explored alternative data sources like SongBPM.com and MusicBrainz. Unfortunately, these sites had outdated information and provided a very limited set of features, such as BPM or key alone, while we required a broader range of audio features for our analysis. Despite these limitations, these sources provided some useful data, though not as comprehensive as the original audio features from Spotify.

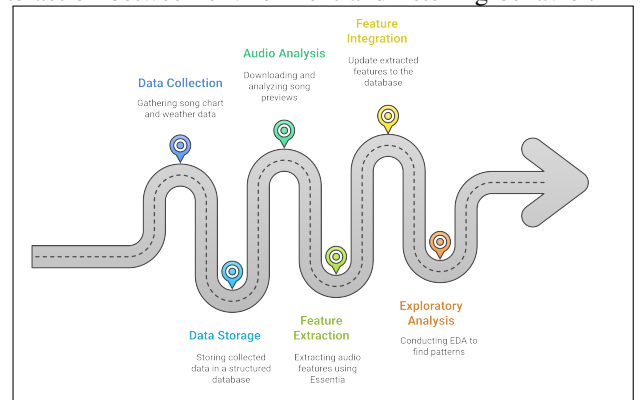
II. RELATED WORK

Several studies have explored the relationship between environmental factors—such as weather and seasons—and music preferences or characteristics.

Anglada-Tort et al. (2023) [7] conducted a large-scale analysis to examine how prevailing weather conditions are reflected in the acoustic features of popular songs. Using a dataset of charting tracks and meteorological data, they found that songs tend to align with environmental moods: for instance, sunny weather was associated with high-energy, positive-valence songs, while cloudy or rainy conditions showed a trend toward slower, lower-energy tracks. Their work suggests that music popularity may be influenced by external atmospheric cues, not just personal taste or cultural factors.

Earlier work by Pettijohn et al. (2010) [8] focused on seasonal music preferences in college students. Their findings showed that individuals tend to prefer slower, more reflective music in winter and upbeat, energetic music during summer, aligning with psychological theories of mood regulation. This research supported the idea that seasonal shifts could influence not only musical consumption but also emotional states through music.

Building on this, Krause and North (2018) [9] explored how listeners create and select seasonal playlists. They found clear differences in genre, tempo, and lyrical content between playlists curated for different seasons. For example, spring and summer playlists included more dance and pop tracks, while autumn and winter featured more acoustic and mellow selections. Their work emphasized the intentional curation of music based on seasonal context, highlighting the dynamic interaction between environment and listening behavior.



Data Pipeline

In 2017, Spotify and AccuWeather launched Climatune [10], a platform that analyzed 85 billion anonymized Spotify streams across over 900 cities to explore the relationship between weather conditions and music preferences. The study found that sunny days were associated with higher-energy, happier-sounding music, while rainy days correlated with lower-energy, more acoustic tracks. However, Climatune was eventually discontinued. The reasons for its shutdown were not publicly disclosed.

Together, these studies touched based trying to derive a correlation between music and environmental context, motivating further exploration of how weather conditions may shape musical preferences or the popularity of specific musical traits which we attempt to do.

III. SOLUTION AND SCALABILITY

A. Solution Description

After overcoming numerous data access and standardization challenges, our final solution involved creating a streamlined, automated pipeline to integrate city-level music chart data with local weather metrics.

The core components of the solution are outlined below:

- **Data Collection and Storage:**
 - *Music Chart Data:* We used Python and Selenium to scrape daily city-level charts from Apple Music, as only current charts are publicly available. The scraped data, including song title, artist, chart rank, and date, was saved in CSV format.
 - *Weather Data:* Simultaneously, weather information (e.g., temperature, humidity, precipitation) was collected using the OpenWeather API and stored daily in CSV files for each city.
- **Database Integration:** A Python script merged the CSV files and imported them into a PostgreSQL database. We structured the data into three relational tables shown in Figure 1:
 - *songs* – Containing unique Spotify song IDs and associated audio features.
 - *charts* – Mapping songs to chart performance by city and date.
 - *weather* – Storing daily weather conditions per city.
- **Audio Feature Enrichment:**
 - Where available, we used Kaggle and other data sources to retrieve audio features such as energy, tempo, and danceability.
 - For songs missing metadata, 30-second preview clips were legally downloaded from Deezer.
 - These previews were analyzed using **Essentia** [11], an open-source audio analysis toolkit developed by the Music Technology Group at Universitat Pompeu Fabra, Barcelona, to extract features like BPM, chord progression, and energy.
 - Additional APIs (e.g., SongBPM, MusicBrainz) were selectively used to fill in gaps.

- **Data Analysis:** Using Python (with `pandas`, `plotly`, and `scikit`), we ran correlation analyses between weather patterns and musical feature trends. Scripts interfaced directly with the PostgreSQL database to query and process data efficiently.

B. Scalability

Scalability was built into the architecture to support growing datasets and increased geographic or temporal scope. Key aspects include:

- **Modular Design:** Each pipeline component (scraping, storage, analysis) can run independently or be scheduled with tools like `cron`.
- **Multithreaded Processing:** The Python script used for Essentia-based audio analysis was multithreaded, dynamically allocating threads based on processor count, significantly reducing runtime for large volumes of audio data.
- **Robust Database Backend:** PostgreSQL supports large-scale relational data storage and optimized querying. Indexing and foreign key relationships allow for rapid joins across songs, weather, and chart data. The system is compatible with cloud-hosted database services for horizontal scalability.
- **Extensibility:** The architecture allows for easy addition of new cities, weather parameters, or audio features. Future integrations may include new streaming platforms or real-time sentiment data from social media.
- **Temporal Expansion:** Though we began with real-time daily data due to historical access limitations, the pipeline is designed for ongoing operation, allowing the dataset to grow continuously for long-term trend analysis.

IV. IMPLEMENTATION DETAILS AND TESTING ENVIRONMENT

A. Software and Libraries Used

The implementation of this project involved using a variety of tools, libraries, and an appropriate testing environment. These tools were essential for data collection, processing, storage, analysis, and testing. The primary components are as follows:

1) *Programming Language and Environment:* The project was developed in Python 3.8 using major libraries listed below, `postgre sql` and `excel`. The environment was managed using the `dotenv` module.

2) *Libraries and APIs:* Several libraries and APIs were used throughout the implementation and testing phases of the project:

- **Selenium:** Used for scraping Apple Music charts by city. Selenium automates a web browser to extract city-level music chart data directly from the Apple Music website.
- **Essentia:** A library for music analysis, used for extracting audio features (e.g., danceability, BPM, energy) from the 30-second previews provided by Deezer. Essentia played a key role in analyzing the music characteristics.

- **Deezer API:** Used to fetch song previews legally, which were crucial for audio feature extraction without violating copyright laws.
- **Spotify API:** Used to fetch Spotify IDs and ensure data consistency by cross-referencing songs and obtaining additional audio features (e.g., loudness, tempo, etc.).
- **pandas:** Used for data manipulation and processing the raw CSV files containing music chart and weather data, as well as merging datasets for analysis.
- **matplotlib** and **plotly:** Utilized for data visualization and generating insightful charts that display trends and patterns in the dataset.
- **logging:** Implemented a logging system for tracking errors, monitoring progress, and debugging the web scraping and data processing steps.
- **PostgreSQL Library for Python:** A relational database management system used to store the music charts, songs, and weather data. The data is stored across three primary tables: charts, songs, and weather.

3) *Database Schema:* The data is organized and stored in a PostgreSQL database, using the following schema as in Figure 1:



Fig. 1. Database Schema

- **Charts Table:** Stores the rankings for each song in various cities, referencing the song's Spotify ID. Each entry contains the city, date, and rank.
- **Songs Table:** Contains metadata about the songs, such as the `spotify_id`, title, artist, album, and various audio features like `danceability`, `bpm`, `energy`, and `valence`.
- **Weather Table:** Stores daily weather data for each city, including temperature, weather conditions, humidity, wind speed, and precipitation.

B. Data Processing and Analysis Workflow

The project followed a structured workflow, which included the following steps:

- 1) **Data Collection:**
 - Scraping city-level music chart data from Apple Music using Selenium.
 - Collecting daily weather data from OpenWeather API.
- 2) **Data Storage:**
 - Storing the scraped data and weather data in CSV format initially.

- Using Python scripts to load the CSV data into PostgreSQL tables.

3) Audio Feature Extraction:

- Using Deezer API to get 30-second song previews and analyzing them using Essentia to extract audio features like BPM, danceability, energy, etc.

4) Correlation Analysis:

- Using Python to perform correlation analysis between weather data and music chart rankings.
- Analyzing how weather conditions relate to the popularity of specific songs or genres.

C. Testing Environment

The project was tested in the following environment to ensure its correct functioning and efficiency:

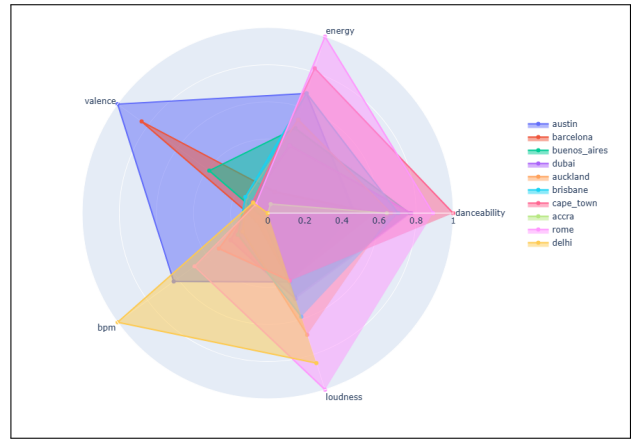


Fig. 2. Music Preferences by Top Cities (Normalized Features)

1) *Hardware Specifications:* The testing was done on a system with the following hardware configuration:

- **Processor:** Intel i7-1185G7 with 4 cores and 16 threads.
- **Memory:** 16GB DDR4 RAM.
- **Storage:** 1TB SSD.
- **Operating System:** Windows 11 chosen for its easy use, however docker was used to run essentia as it is linux only

2) *Libraries and Tools Used in Testing:* For testing purposes, the following libraries and tools were utilized:

- **Python Version:** Python 3.8 within a virtual environment.
- **Database:** PostgreSQL 13 was used locally for database operations.
- **Libraries:** unittest for unit testing, pandas for data handling, and logging for error tracking.
- **Tools:** Docker for running essentia as it is linux only

V. RESULTS AND DISCUSSIONS

As part of our analysis, we conducted Exploratory Data Analysis(EDA) to identify potential correlations between weather conditions and music preferences. We used the audio features of Beats Per Minute(BPM), energy, danceability,

loudness, chords, genre and valence in addition to weather attributes of temperature and precipitation. It is important to note that these findings are indicative of a direction rather than conclusive given the complexity of human behavior and multi-dimensional nature of the data we have.

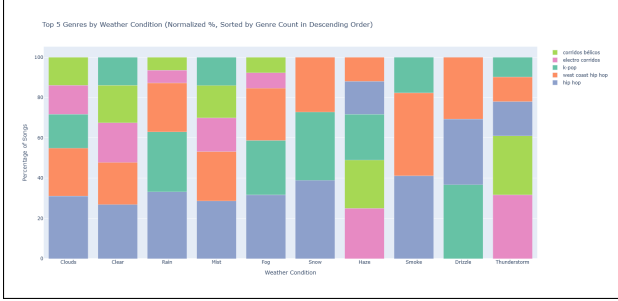


Fig. 3. Top 5 Genres By Weather (Normalized & Sorted in Descending Order)

Figure 2 presents a radar chart showing cities with the highest and lowest average values across selected audio features. For example, *Austin* exhibits the highest **valence** (a measure of musical positivity), suggesting a general preference for more upbeat music. In contrast, *Delhi* shows the highest average **BPM** (beats per minute), reflecting a tendency toward faster-paced tracks, which aligns with the prevalence of higher beats music in Indian culture. These patterns offer a glimpse into how regional culture and local music trends may shape listening behaviors.

Figure 3 illustrates the top five most common music genres listened to across all cities during the data collection period. The data has been normalized by weather condition to highlight relative preferences. Notably, *hip hop* and *west coast hip hop* appear consistently across all weather types, suggesting a stable preference for these genres. In contrast, *K-pop* and *electro corridos* show significant spikes during *haze* and *thunderstorms*, indicating that dramatic weather conditions might influence mood-related listening behavior. Overall, while some genres fluctuate with weather, *hip hop* remains a constant, dominant choice regardless of external conditions.

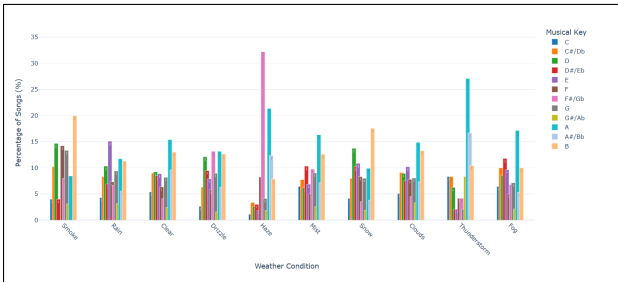


Fig. 4. Musical Key Distribution by Weather

Figure 4 chart reveals a non-uniform distribution of musical keys across diverse weather conditions, suggesting a potential connection between atmospheric states and musical choices. Key A remains quite consistent across many weather types

like Mist, Clouds, and thunderstorms, making it one of the more stable and commonly used keys overall. One interesting thing to note is that Key B stands out during Smoke and Snow, possibly pointing to a preference for deeper or more intense tones in those conditions. Whereas a very noticeable spike is seen for F-sharp or G-flat during Haze, which could suggest a unique pattern or even an anomaly in musical mood under hazy weather. This gives us a meaningful insight into how weather could be subtly shape our music choices.

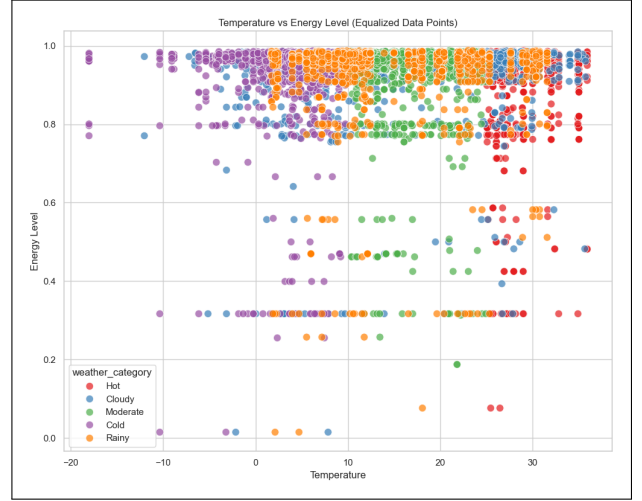


Fig. 5. Temperature vs Energy (Equal Data Points)

Figure 5 shows a scatter plot showcasing the relationship between temperature and music energy levels, categorized by weather conditions. There is a slight trend indicating that in warmer conditions—particularly under "Hot" and "Cloudy" categories—listeners tend to prefer higher-energy music. However, it's important to acknowledge that the dataset has a high average energy level of **0.93**, likely due to the use of song previews, that typically highlight the most dynamic parts of a track. This introduces a potential bias, inflating overall energy values. While the plot suggests a correlation between warmer temperatures and higher energy levels, the evidence is not strong enough to draw definitive conclusions. That said, the general pattern aligns with intuitive expectations about how weather may influence musical energy preferences.

Figure 6 presents a 2D visualization of city clusters based on average music features—*BPM*, *energy*, and *valence*—along with average temperatures recorded on days above 0°C. These features were standardized and reduced to two principal components using Principal Component Analysis (PCA) [12], where *PCA 1* and *PCA 2* represent the directions of highest variance. Each point on the plot corresponds to a city, and color-coded KMeans clusters (with $k = 4$) group together cities exhibiting similar listening patterns under comparable weather conditions. While $k = 4$ was selected for ease of interpretation, alternative values could yield different cluster configurations.

From the visualization, we observe the following trends:

- **Cluster 0 (green)** includes many European and East Asian cities (e.g., London, Tokyo, Berlin), reflecting moderate and balanced musical preferences.
- **Cluster 1 (orange)** includes cities like Mumbai, Manila, Lima, and Santiago, marked by distinctive listening patterns potentially driven by hot and humid climates that favor upbeat or rhythmic music.
- **Cluster 2 (blue)** comprises cities such as Lagos, Johannesburg, and Rio de Janeiro, which tend toward higher *BPM* and *energy*, particularly in warmer climates.
- **Cluster 3 (pink)** groups North and Latin American cities including Dallas, Monterrey, and Los Angeles, suggesting a preference for high-energy and valence-rich tracks.

This clustering approach reveals how environmental and cultural contexts can shape regional music consumption behavior. It also demonstrates the utility of combining audio and weather data to inform music recommendation systems, targeted marketing strategies, and cross-cultural studies in digital music trends.

Overall, the exploratory analyses highlights nuanced relationships between environmental conditions and music listening behavior. While some trends, such as the consistent popularity of hip hop or the preference for major keys during clear weather, appear stable, others suggest situational influences, such as spikes in K-pop during thunderstorms or elevated BPM in hotter regions like Delhi. The PCA-based clustering further underscores that musical preferences are not only culturally rooted but also environmentally influenced. These insights, while indicative, underscore the need for deeper modeling and broader data to draw inferences.



Fig. 6. City Clusters by applying PCA to BPM, Energy, Valence and Temperature

VI. CONCLUSION & FUTURE WORK

This study explored the intersection of music listening behavior and weather patterns by analyzing genre preferences, audio features, and city-level listening trends using data from charts from streaming platforms. While consistent patterns emerged—such as the widespread popularity of hip hop regardless of weather, or the correlation between warm climates and higher-energy music—many findings suggest contextual influences tied to culture, climate, and temporal factors. The

use of previews in our dataset introduced a bias toward higher-energy songs, highlighting the need for more representative samples in future analyses. Our clustering results demonstrated that regional listening behavior is not only culturally specific but also subtly influenced by environmental conditions.

For future work, incorporating a wider array of features such as user demographics, time-of-day patterns, and additional weather variables like humidity or wind speed could yield more robust models. Deep learning techniques or time-series modeling could also uncover more dependencies in music preferences. Additionally, collecting full-track audio data rather than previews, and integrating cross-platform behavior (e.g., YouTube Music, Apple Music) would provide a more comprehensive view of how music consumption reflects and reacts to external stimuli. Ultimately, this line of research opens avenues not just in music recommendation systems, but also in understanding mood, cultural behavior, and environmental psychology at a global scale.

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